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Sport Finance and Economics

Portfolio Risk and Diversification

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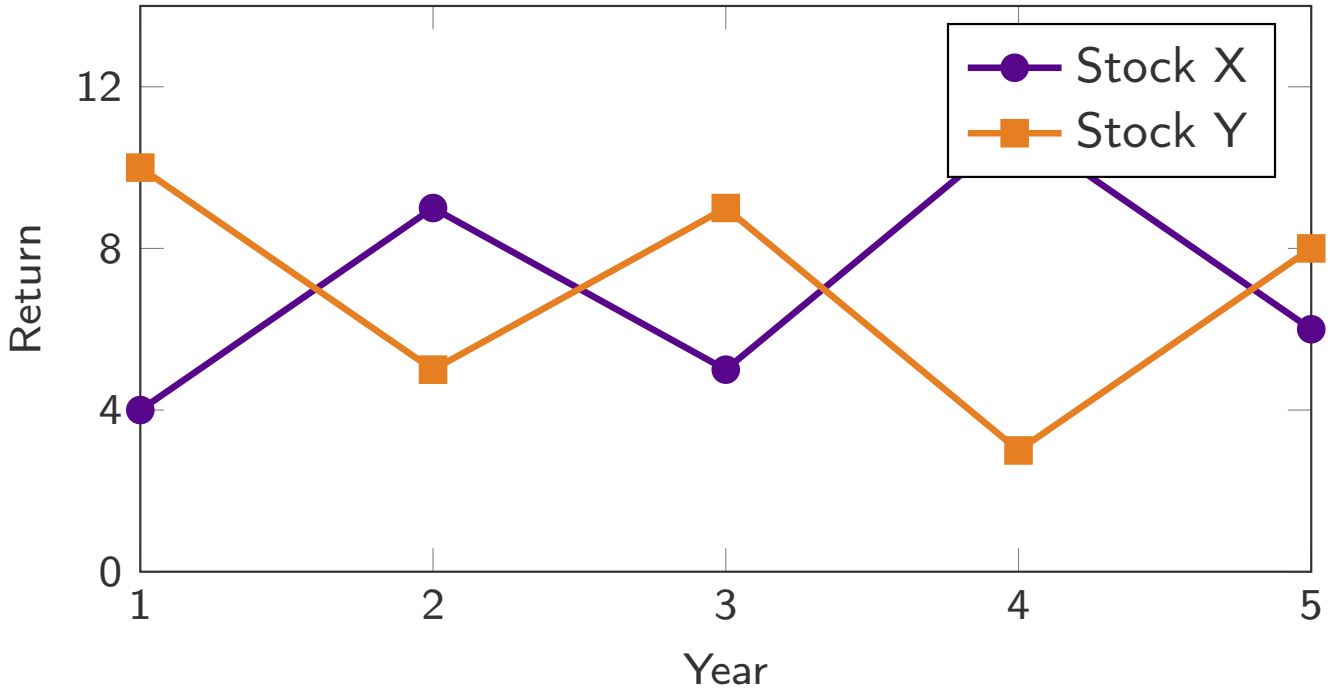
Why portfolio risk is not just a weighted average

- Portfolio risk depends on how assets move relative to one another.
- Adding a new asset can reduce risk if its returns are not perfectly positively correlated with the current portfolio.
- Diversification reduces some risk, but not all risk.

Correlation coefficient

$\sigma_{xy} = +1$: assets move together perfectly. $\sigma_{xy} = 0$: no linear relation. $\sigma_{xy} = -1$: assets move in opposite directions perfectly.

Correlation intuition



Key takeaway

The risk reduction comes from imperfect co-movement. When one asset performs poorly, another may cushion the portfolio.

Exhibit 3.7 in brown2026chapter.

Portfolio expected return and risk

$$E(r_p) = \mathrm{w}^\top \boldsymbol{\mu} = \sum_{i=1}^{n} w_i E(r_i)$$

$$\sigma_p^2 = \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w} = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \, \mathrm{Cov}(r_i, r_j)$$

$$\sigma_p = \sqrt{\mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}}, \qquad 1^\top \mathrm{w} = 1$$

r_p : portfolio return

w : $n \times 1$ vector of portfolio weights

$\boldsymbol{\mu}$: $n \times 1$ vector of expected asset returns

$\boldsymbol{\Sigma}$: $n \times n$ covariance matrix

$\sigma_p^2, \ \sigma_p$: portfolio variance and standard deviation

n : number of assets

Constructing the Markowitz efficient frontier

$$\begin{aligned} \min_{w} \quad & w^\top \mathbf{\Sigma} w \\ \text{subject to} \quad & w^\top \boldsymbol{\mu} = \mu^*, \\ & 1^\top w = 1. \end{aligned}$$

$$w_{GMV} = \frac{\mathbf{\Sigma}^{-1} 1}{1^\top \mathbf{\Sigma}^{-1} 1}, \qquad \sigma^2_{GMV} = \frac{1}{1^\top \mathbf{\Sigma}^{-1} 1}$$

μ^* : target expected portfolio return

w_{GMV} : global minimum-variance portfolio weights

σ^2_{GMV} : global minimum portfolio variance

$\mathbf{\Sigma}^{-1}$: inverse covariance matrix

1 : $n \times 1$identity vector

Constructing the tangency portfolio

$$w_M = \arg\max_w \frac{w^\top \boldsymbol{\mu} - r_f}{\sqrt{w^\top \boldsymbol{\Sigma} w}}$$

$$w_M = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f 1)}{1^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - r_f 1)}$$

$$E(r_M) = w_M^\top \boldsymbol{\mu}, \qquad \sigma_M = \sqrt{w_M^\top \boldsymbol{\Sigma} w_M}$$

w_M : tangency-portfolio weights

r_f : risk-free rate

$E(r_M)$: expected return of the tangency portfolio

σ_M : standard deviation of the tangency portfolio

$\frac{E(r_M) - r_f}{\sigma_M}$: Sharpe ratio of the tangency portfolio

The Capital Market Line

$$r_p = (1 - y)r_f + yr_M$$

$$E(r_p) = r_f + y\left[E(r_M) - r_f\right], \qquad \sigma_p = y\sigma_M$$

$$E(r_p) = r_f + \frac{E(r_M) - r_f}{\sigma_M}\sigma_p$$

y : proportion invested in the market portfolio

$1 - y$: proportion invested in the risk-free asset

r_M : return on the market portfolio

$E(r_M) - r_f$: market risk premium

σ_M : market-portfolio standard deviation

σ_p : standard deviation of the complete portfolio

$0 < y < 1$: invest y in the market and $1 - y$ in the risk-free asset $\rightarrow$ lending

$y = 1$: invest 100% in the market portfolio

$y > 1$: borrow $y - 1$ at the risk-free rate and invest more than 100% in the market $\rightarrow$ leveraging the portfolio

Diversifiable risk vs. market risk

Diversifiable risk

Firm-specific risk that can be reduced by holding many different assets.

- poor team performance
- local sponsorship loss
- facility-specific issue
- management error

Market risk

Systematic risk that affects most assets and cannot be diversified away.

- recession
- inflation
- interest rates
- broad credit shock

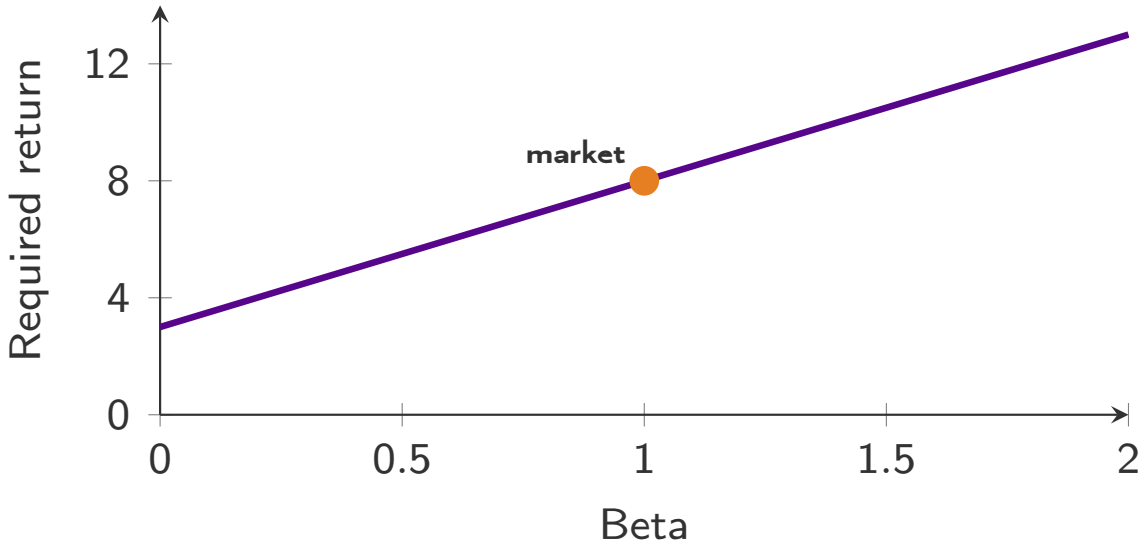
Key takeaway

Under CAPM, the relevant risk is market risk, because diversifiable risk can be reduced in a well-diversified portfolio.

CAPM and the security market line

Required return under CAPM

$$r = r_{RF} + \beta(r_M - r_{RF})$$



Beta coefficient

- Beta measures how sensitive an asset is to movements in the overall market.
- $\beta = 1.0$: moves with the market.
- $\beta > 1.0$: more volatile than the market.
- $\beta < 1.0$: less volatile than the market.

Portfolio beta

$$\beta_p = \sum_{j=1}^{m} w_j \beta_j$$